



ZHAO XIAOTING | PORTFOLIO
CHONGQING UNIVERSITY

PROJECT



A photo in West Bund Museum, Shanghai, David Chipperfield



College Selects Outstanding Assignment
FALLING LEAF
The Boundary of Lightness and Permeability
Design of Student Activity Center and Theater, Huxi Campus, Chongqing University

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Huayi Cup Outstanding Creative Design Award
HANDING ROCK
Urban Transformation Revisited
Supertall Design for Coastal Cities in the Context of Urban Renewal

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International Solar Building Design Competition
FALLING INTO THE VALLEY
Nature-oriented
Design of Scientific Research Station for Nature Conservation Community in Guanba Gou Watershed, Pingwu County, Sichuan Province, China

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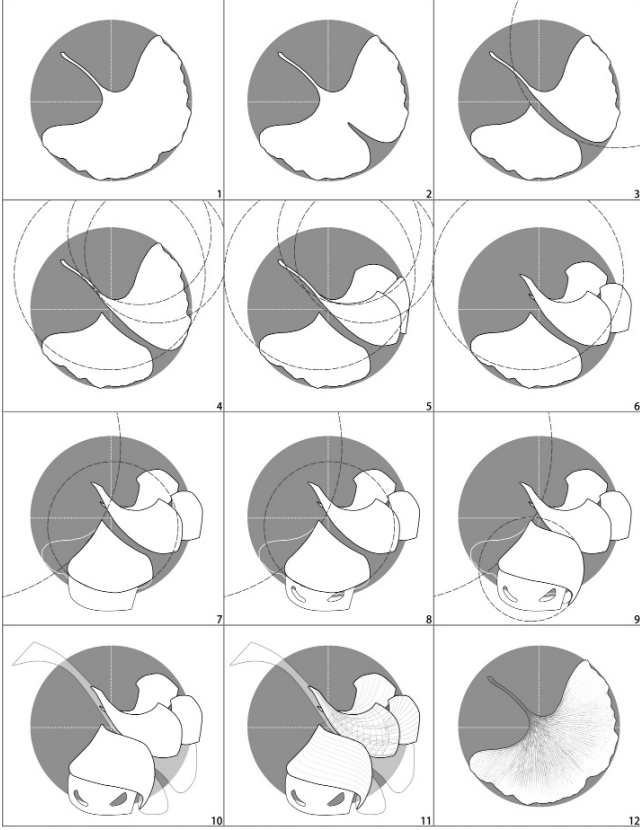


eVolo Skyscraper Competition
ANTARCTIC RESCUE LINE
Mending Antarctica
Antarctic Underground Thermal Energy Conversion Station Design

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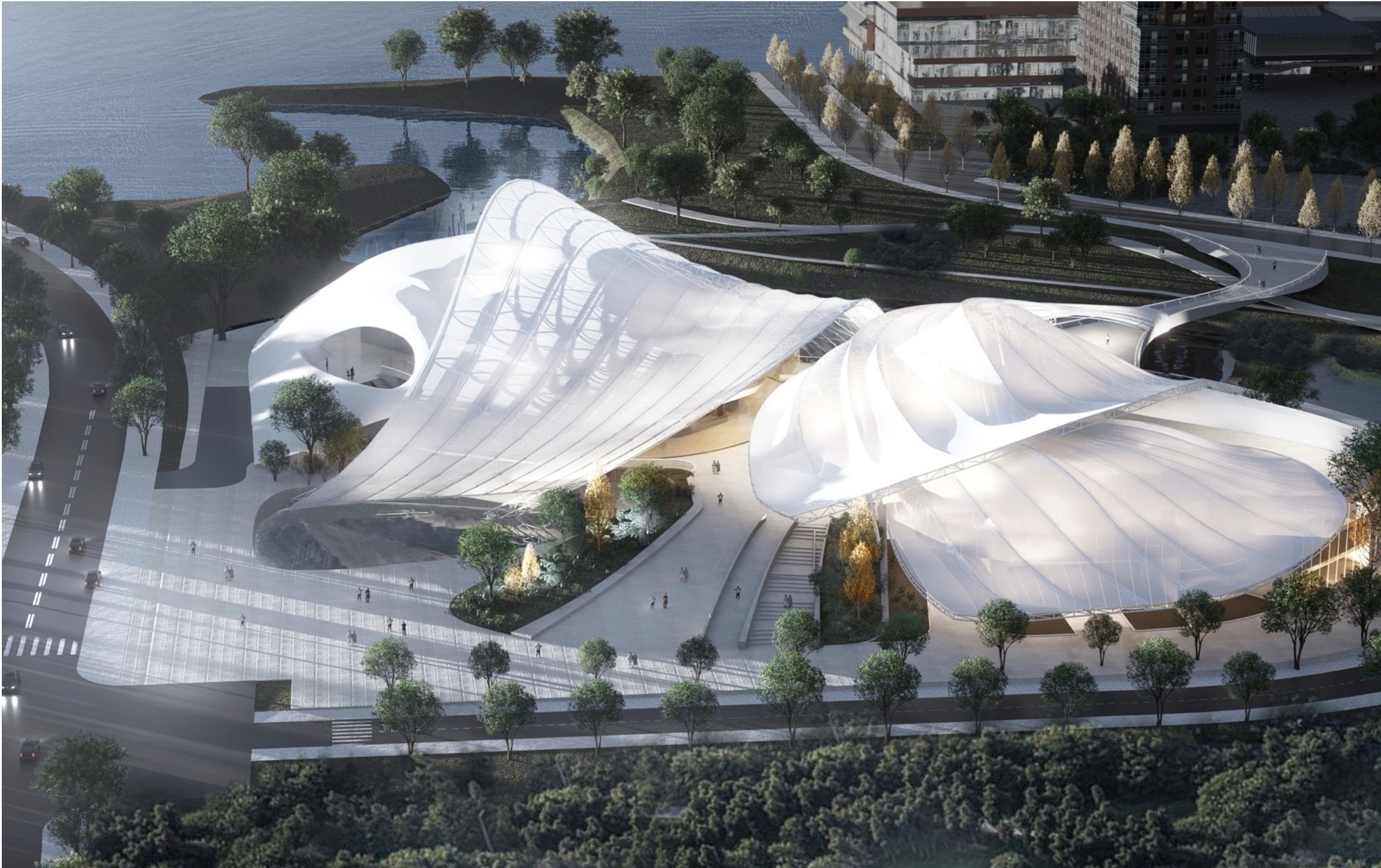
FALLING LEAFS

FORM DIAGRAM

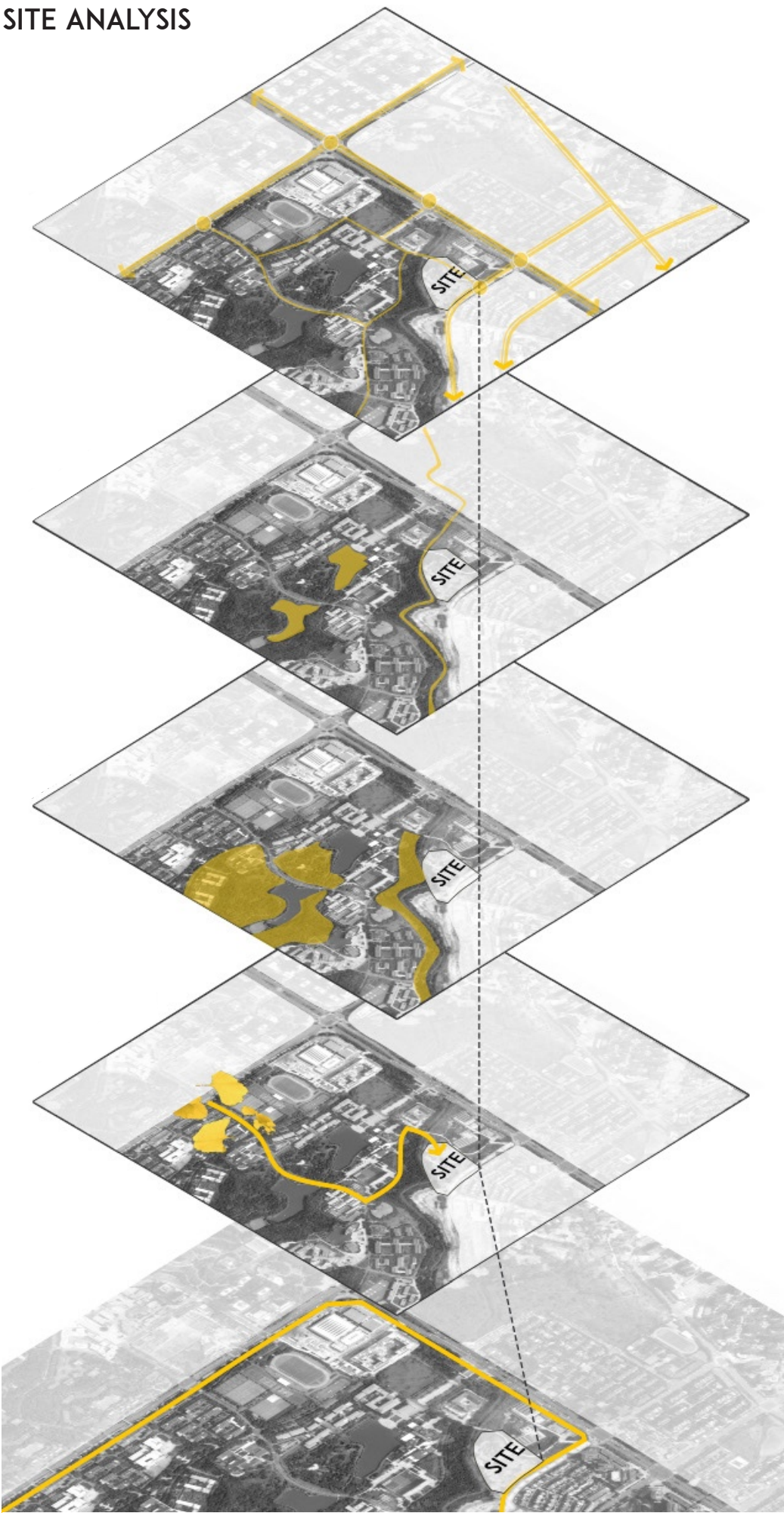


DESIGN DESCRIPTION

The site is located in Huxi Campus of Chongqing University in Shapingba District, Chongqing, which is a triangular-shaped site surrounded by the city's main road, the school road and the Huxi River. The design borrows the important Ginkgo Biloba Boulevard, which starts at the north gate of the campus and ends at the library in the south, and connects it to the site, allowing the continuation of this important cultural icon from north to south through the entire campus.

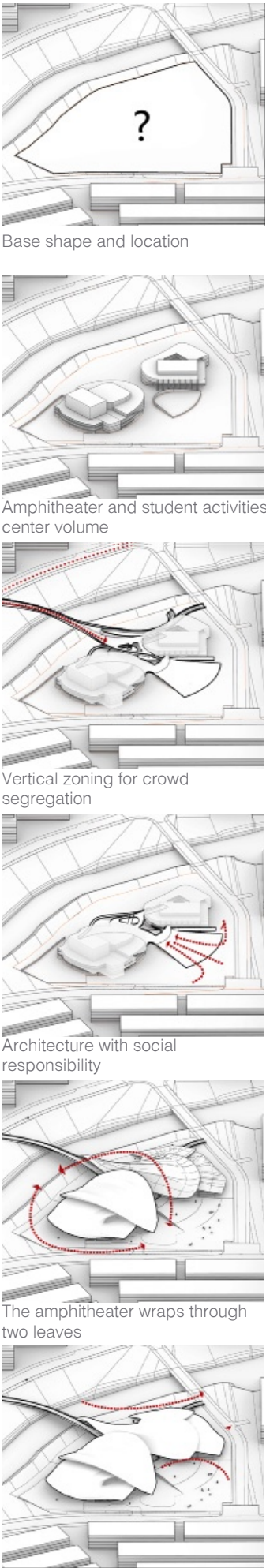


SITE ANALYSIS

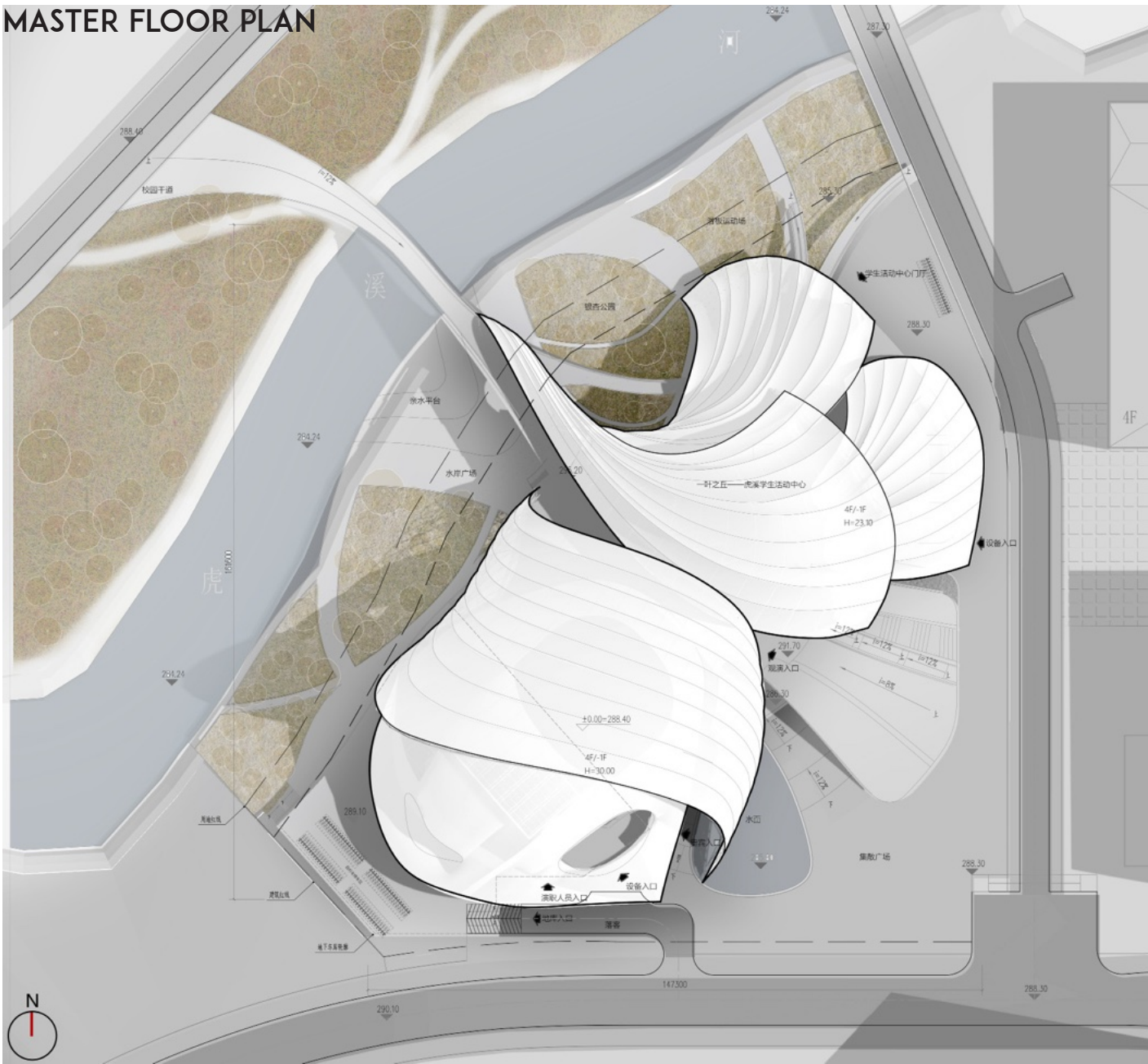


Good landscape resources; good layout and variety; easy access to both sides; complex traffic.

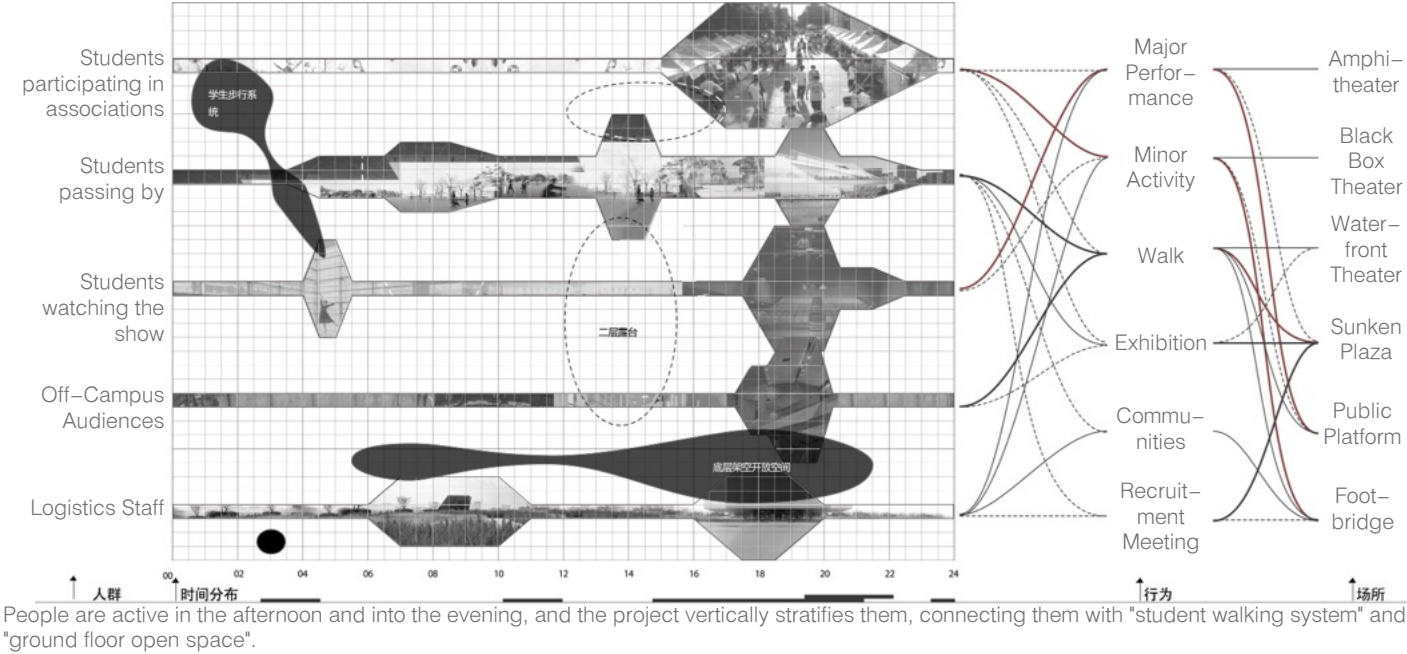
VOLUME

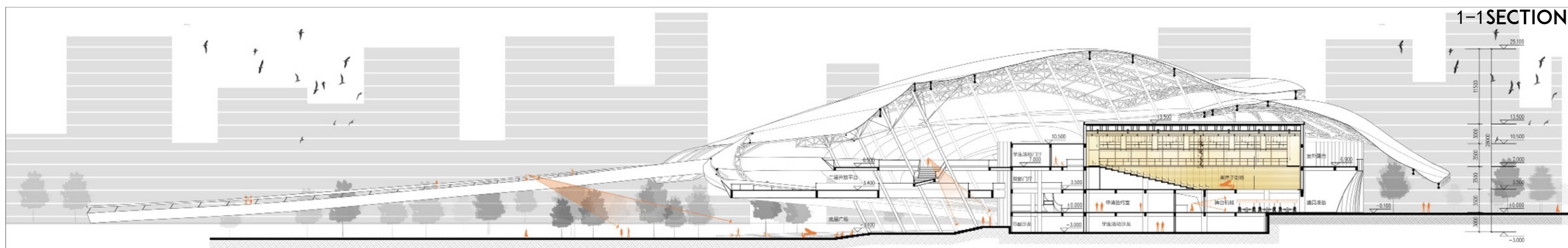


MASTER FLOOR PLAN



VENUES, PEOPLE AND TIME

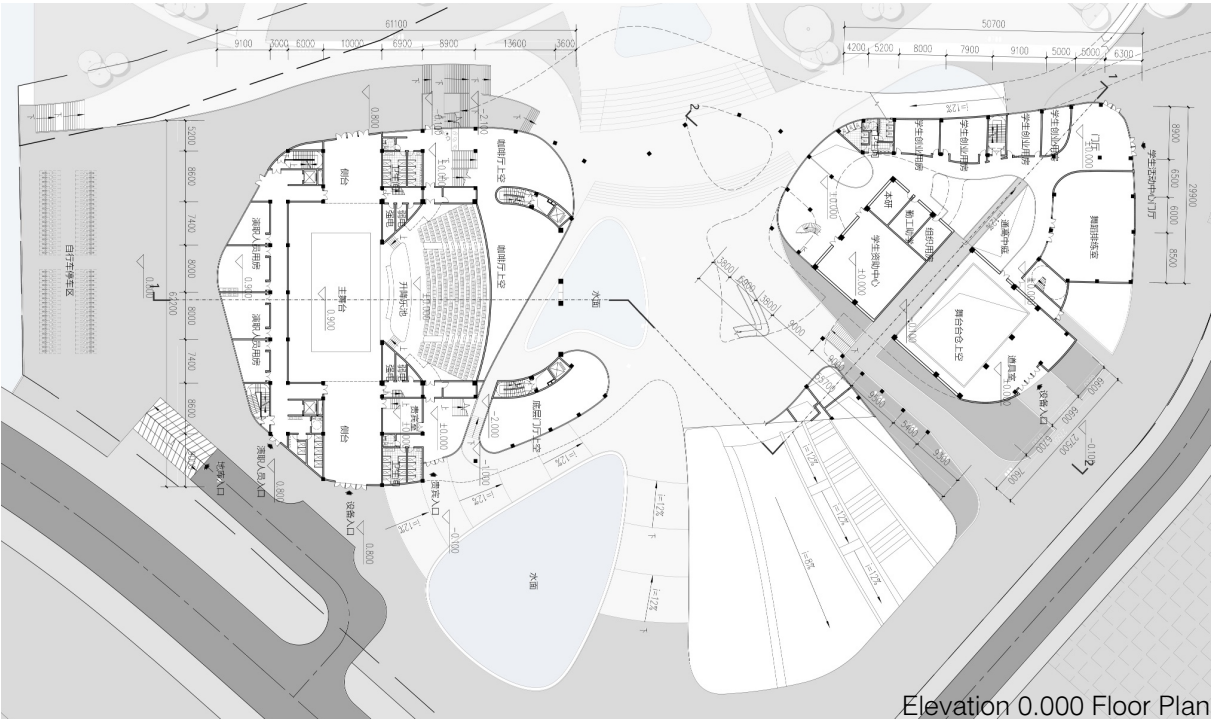




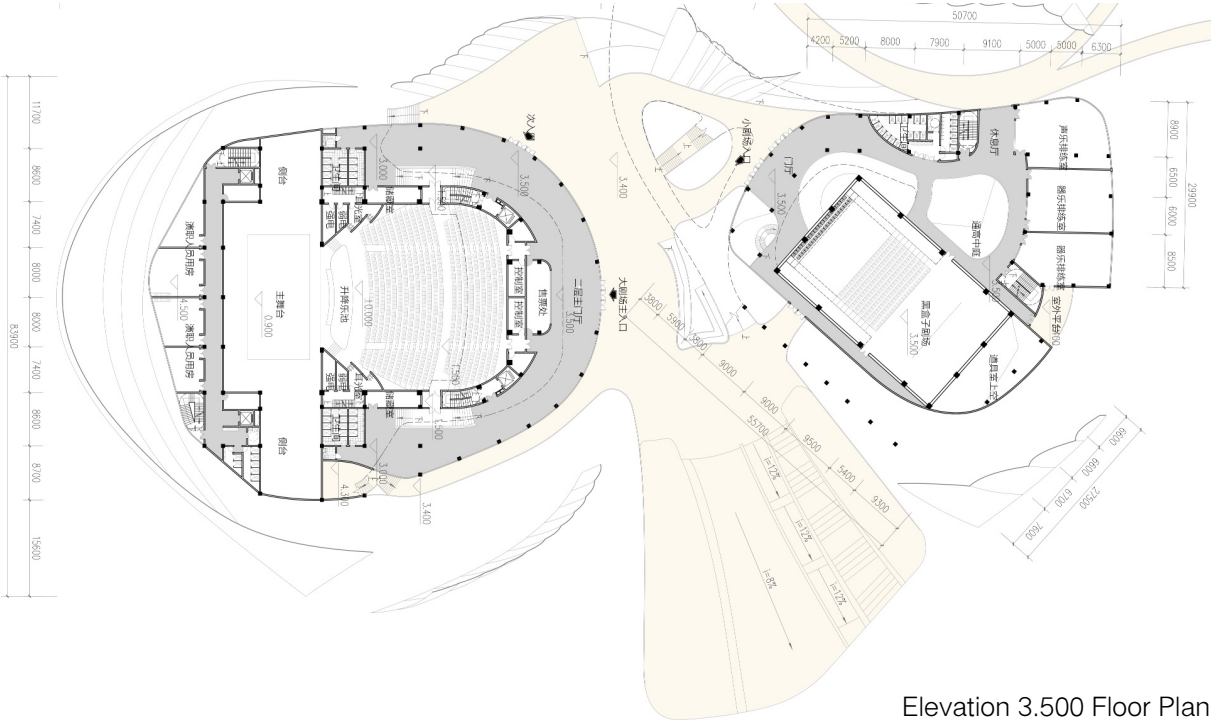
PLANS OF EACH LEVEL



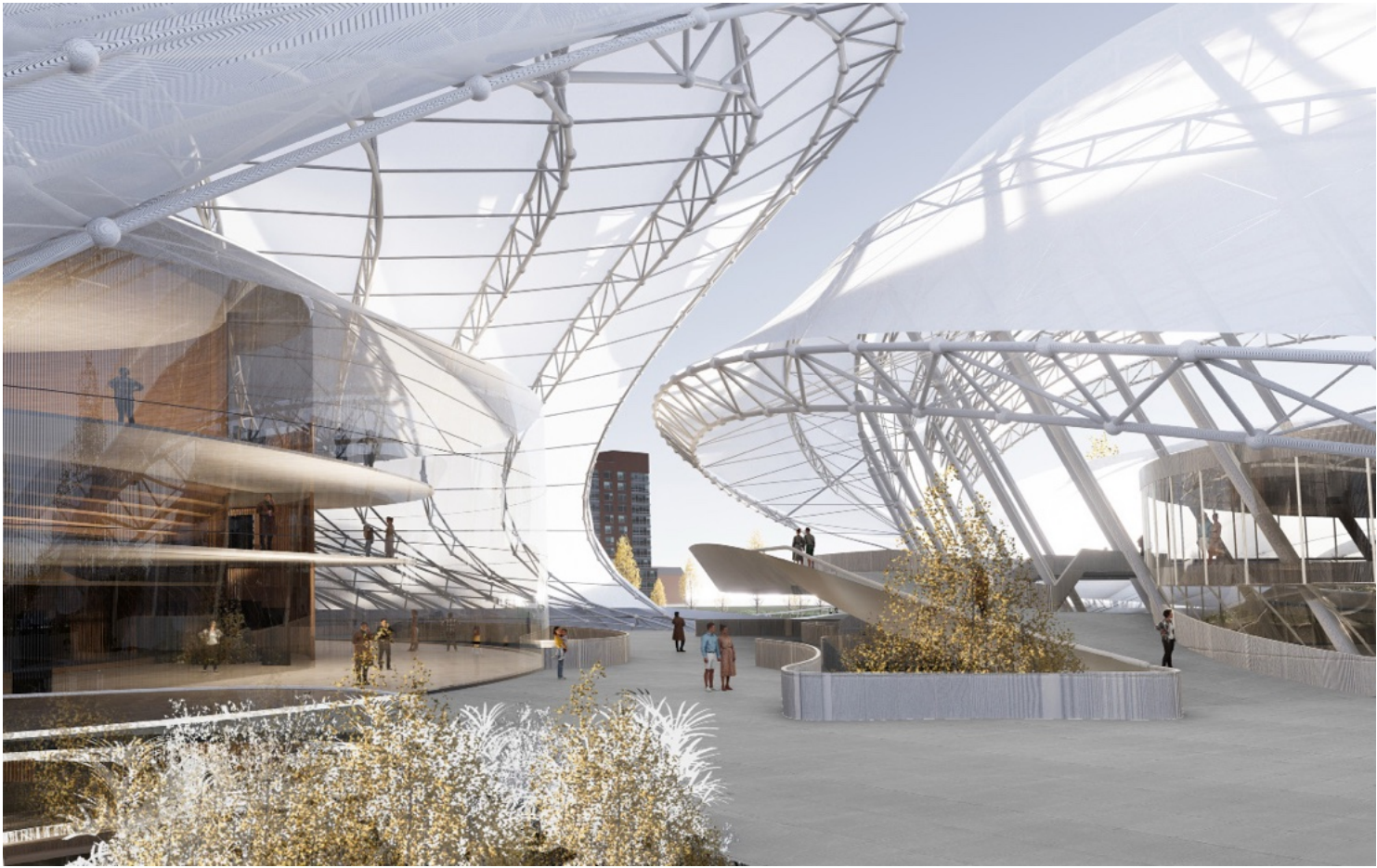
Elevation -2.000 Floor Plan



Elevation 0.000 Floor Plan

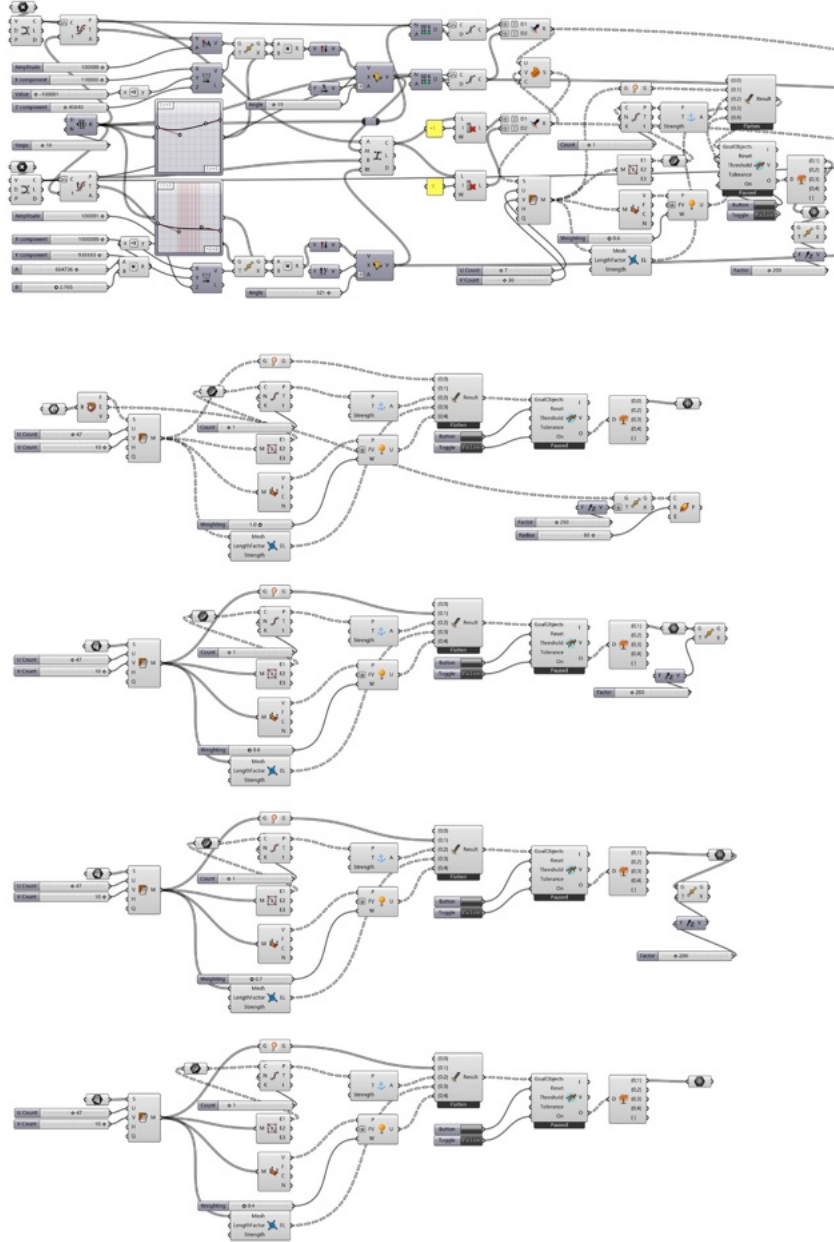


Elevation 3.500 Floor Plan

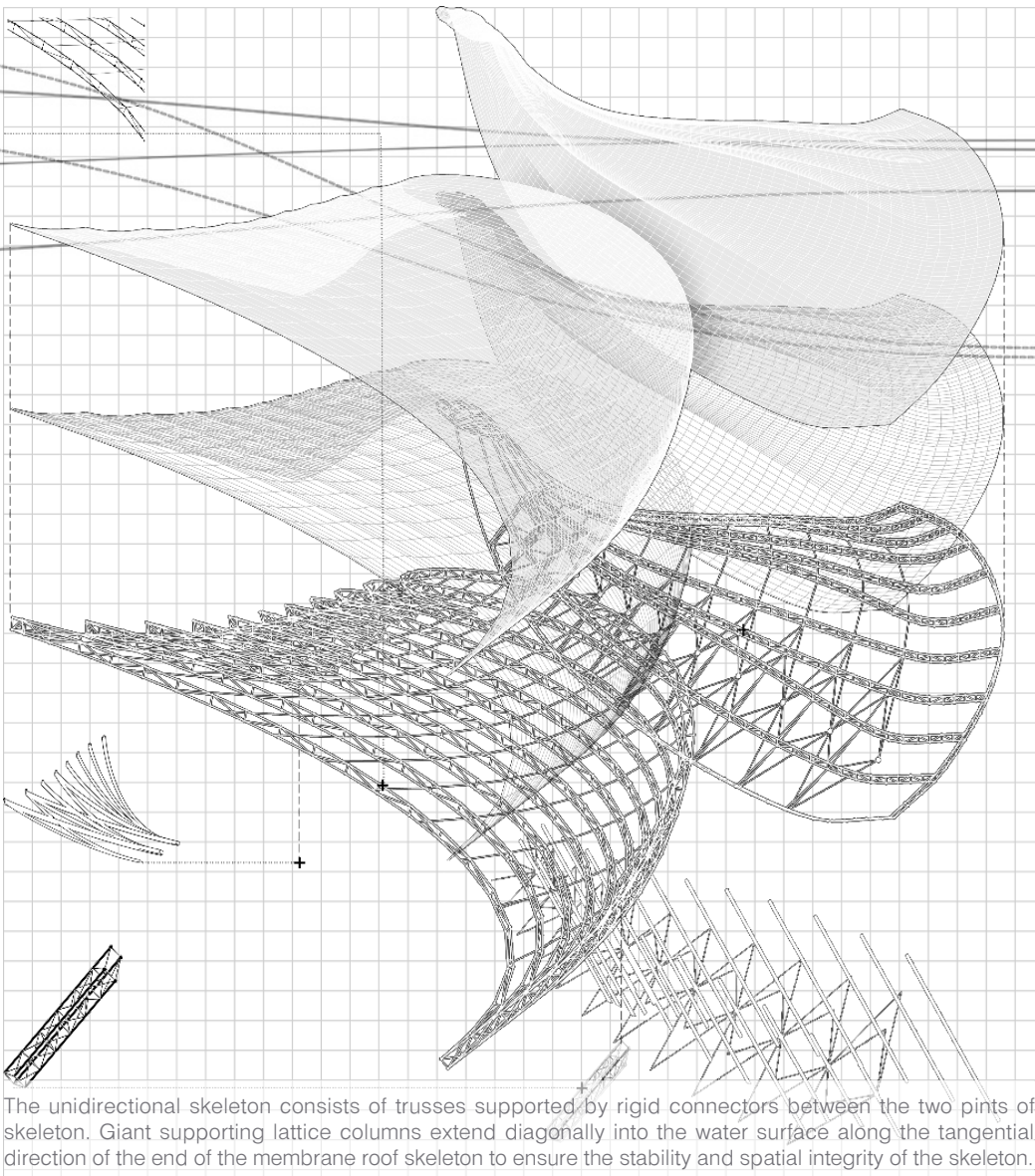


The space keeps folding and unfolding to form a continuous narrative, and then shrinks inward under the action of external forces; the boundaries between the city, the campus, and the natural environment disappear; the whole building dissolves and fades away from its surroundings; and the people who go upward and downward are unobstructed.

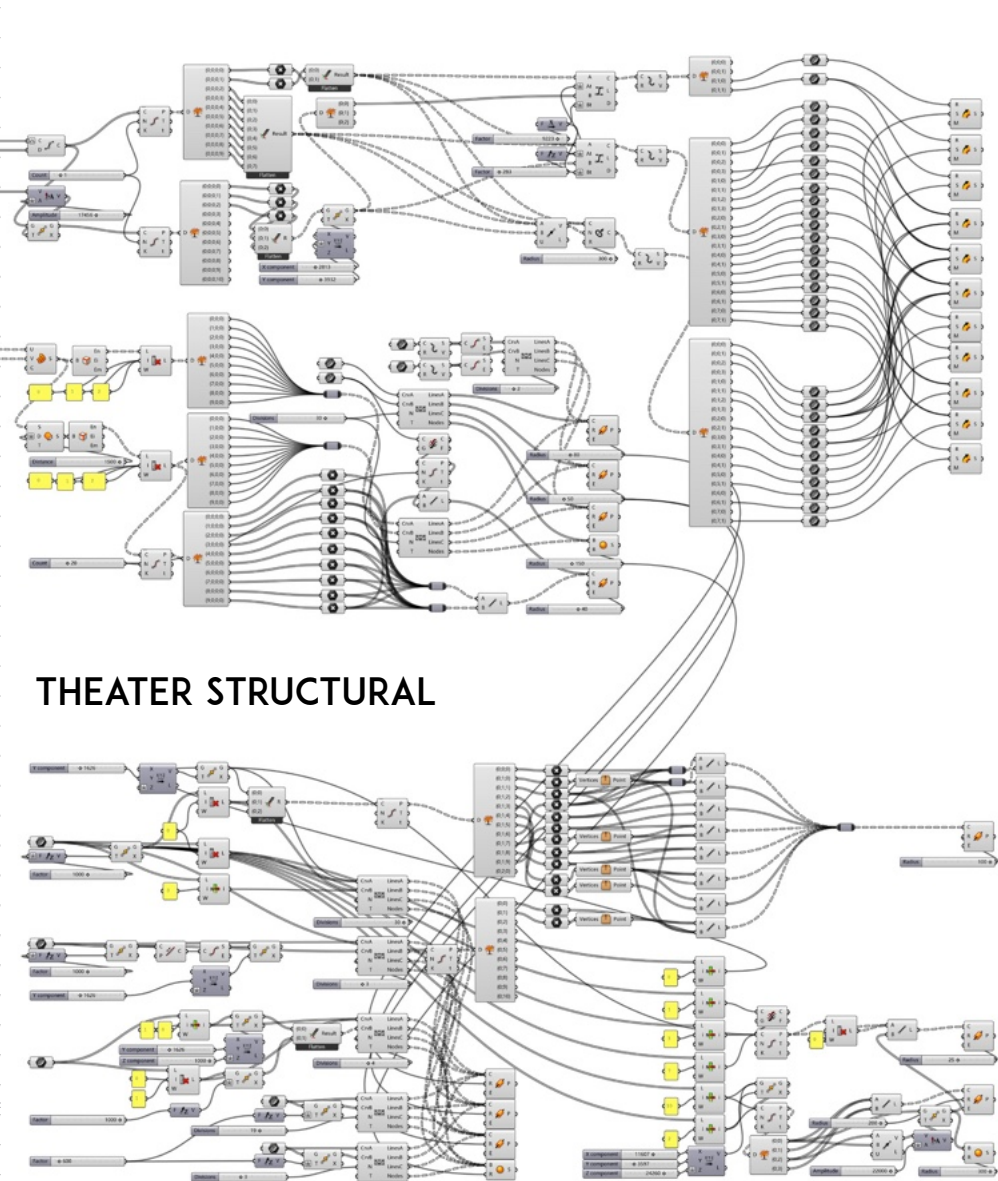
MEMBRANE MATERIAL



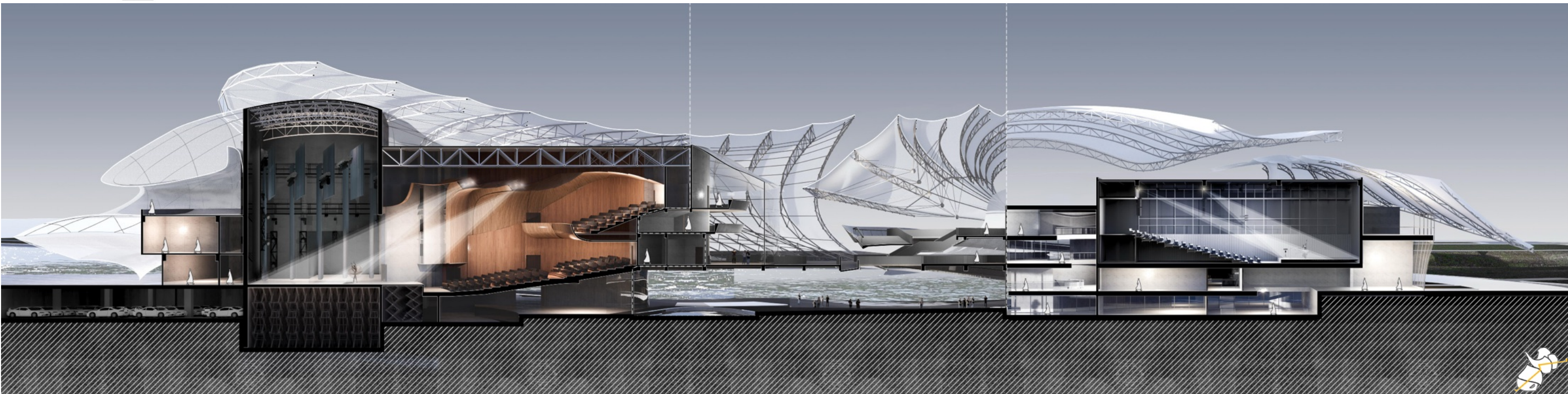
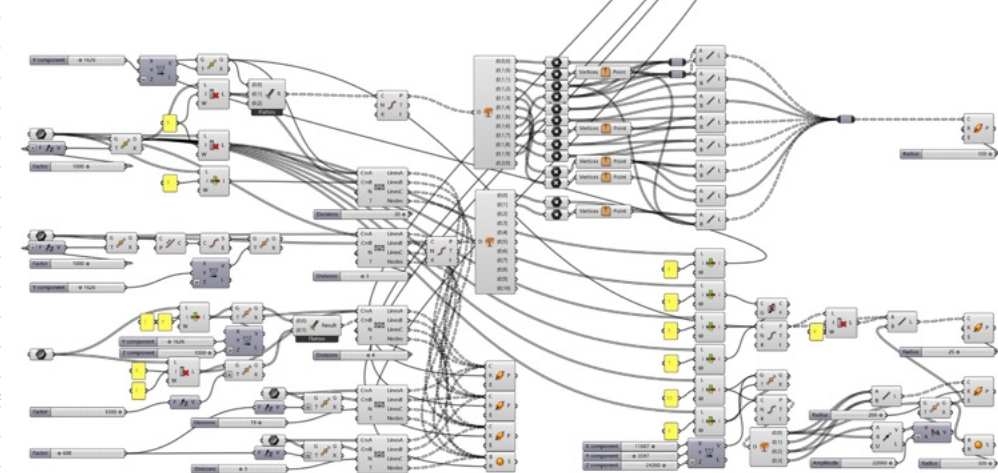
ROOF STRUCTURAL ANALYSIS



ACTIVITY CENTER STRUCTURAL

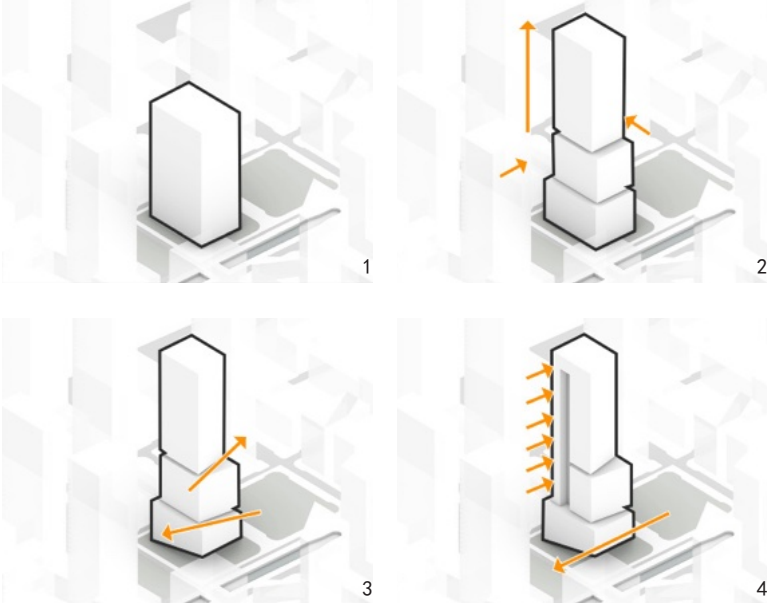


THEATER STRUCTURAL



HANGING ROCK

VOLUME GENERATION



DESIGN DESCRIPTION

Located in the corporate headquarters area of Houhai, Shenzhen, the design starts from the historical development of the coastline of Houhai, Shenzhen, and uses the futuristic Hanging Stones as a source of inspiration, responding to the urban elements and the background of technological enterprises. The design runs through the concept of hanging stones, where the gaps between the stones give way to multi-level outdoor activity platforms. The ground floor blurs the urban boundaries through a cantilevered + fully transparent glass façade, echoing the surrounding green space.

The building intends to use sunlight and air to shape the space, adopting offset cylinders to maximize light surfaces and workplaces, while the north-south green slit facilitates the creation of breezes to reduce energy consumption and contact with nature. The electrically adjustable unitized curtain wall enables ventilation in summer and heat preservation in winter. The design not only shapes a new landmark in Houhai, but also explores the possibilities of future super high-rise buildings.



SITE



The land use of this project is commercial. It is located in the core area of Houhai, Nanshan District.

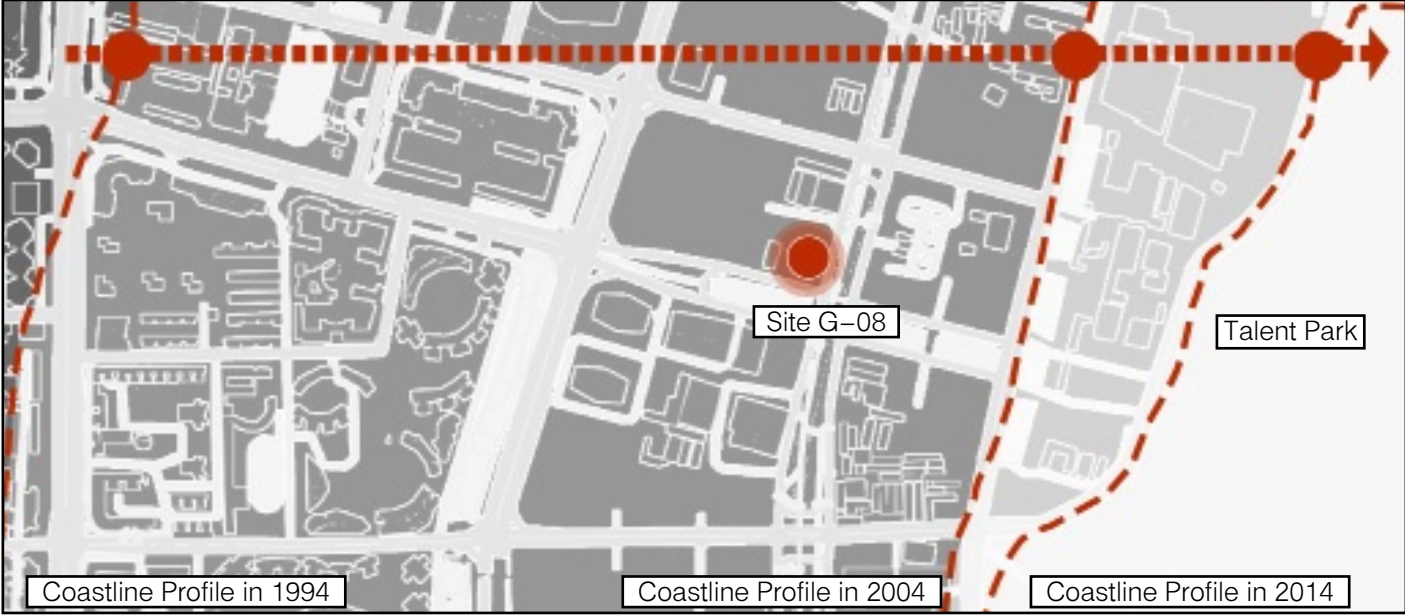
BACKGROUND



The project service object is a leading enterprise in the field of digital science and technology application, which is committed to the research and development of digital new media technology.

Relying on the company's strong creative, design and content production capabilities, combined with the key technical means of CG special effects, human-computer interaction, holograms, etc. accumulated over the years, the company's products and systems are widely used in the fields of urban cultural experience, digital display and digital art consumption.

COASTLINE CHANGE



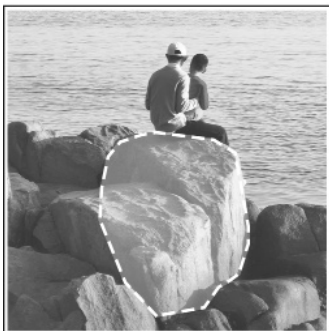
One of the more important studies for our design concept is the historical development of the coastline of Deep Bay. Shenzhen is a city expanding towards the sea, and in 2004 the coastline development was pushed to the front of the site, while in 1994 our site was still submerged in the sea. The site reflects the transformation of the coastal city of Shenzhen, and the concept of the project cuts through this.



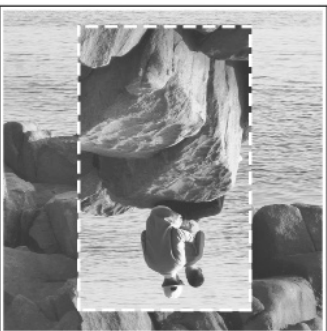
Past State of the Site



Current State of the Site

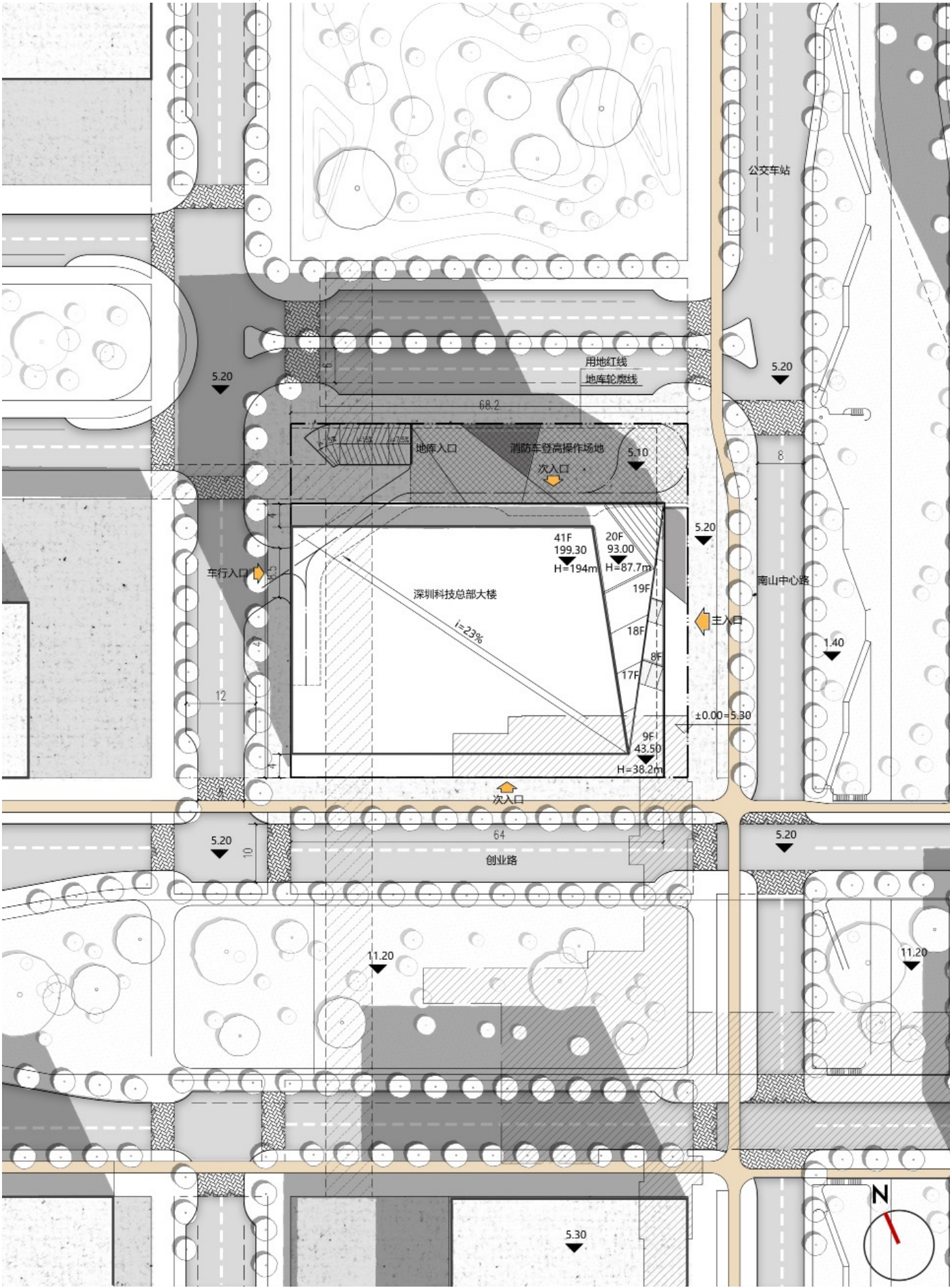


Place Memory in Locality



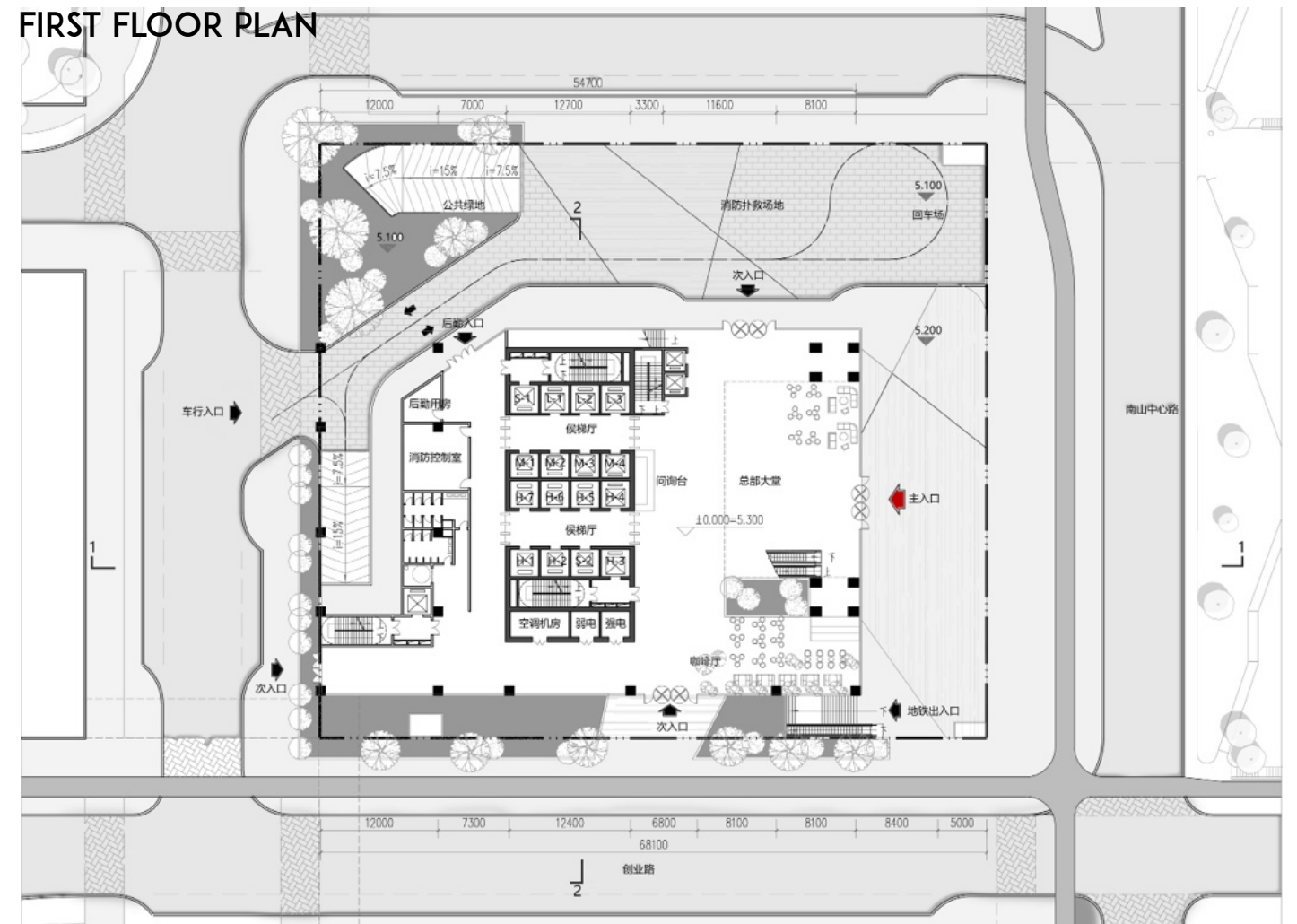
Futuristic Corporate Culture

MASTER FLOOR PLAN





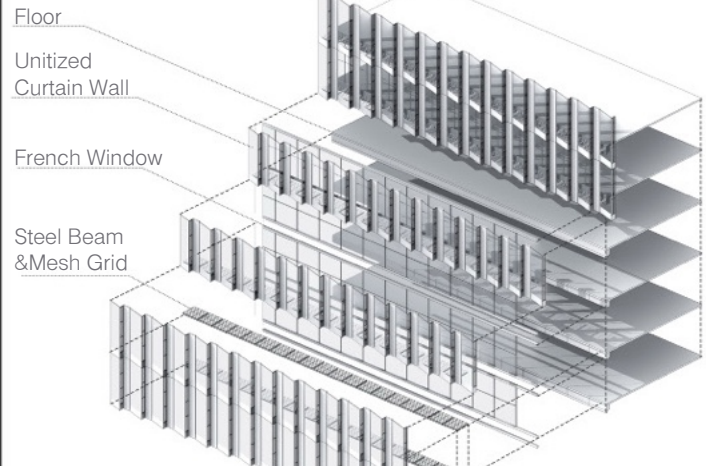
FIRST FLOOR PLAN



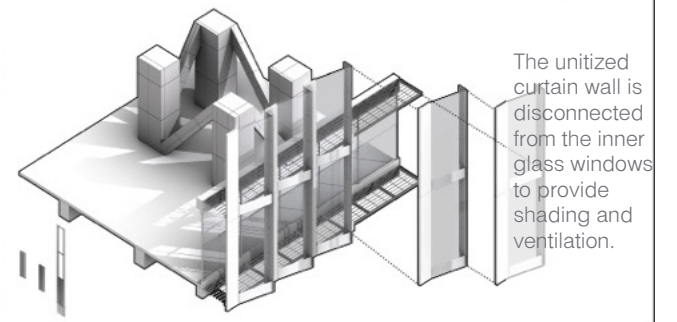
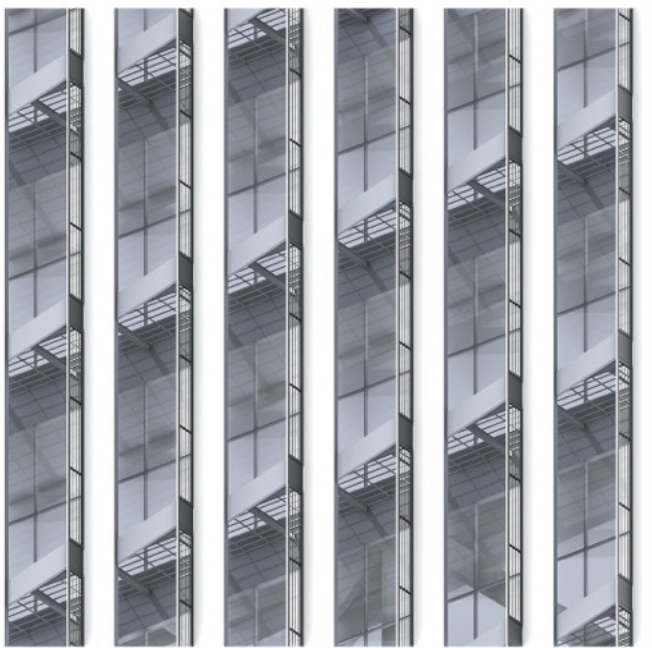
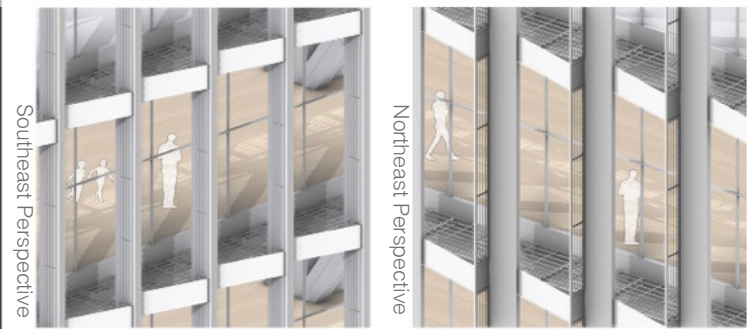
The site conditions are confined. Under the premise of meeting the code, the podium could hardly form a scale, so the podium was eliminated, the bottom floor was increased to meet the functional demand, and the bottom contracted inward to form a huge gray space to connect the north and south crowds with the green corridor. When viewed from the street corner, the lobby and the building boundary form an incomplete view frame.



CURTAIN WALL EXPLODED VIEW

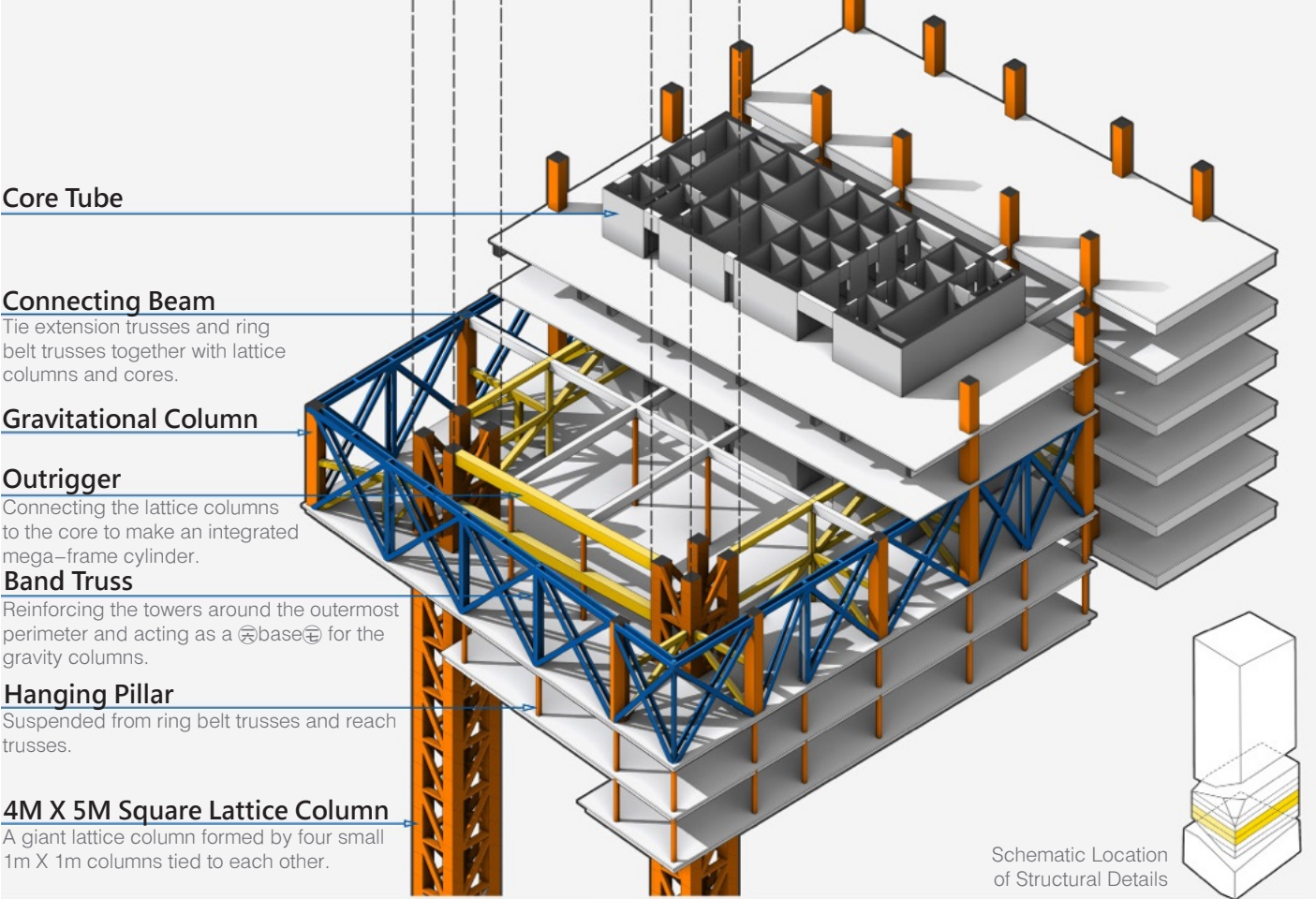


The building utilizes a modular curtain wall that is disconnected from the main body of the building: floor-to-ceiling glazing on the inside, and an adjustable, breathable modular curtain wall on the outside. The curtain wall is connected to the floor slabs by I-beam steel beams, which are covered with metal mesh grids to provide sun shading and reinforce the beams against wind loads.



The unitized curtain wall is disconnected from the inner glass windows to provide shading and ventilation.

LOAD-BEARING CONSTRUCTION EXPLODED VIEW



Core Tube

Connecting Beam

Tie extension trusses and ring belt trusses together with lattice columns and cores.

Gravitational Column

Outrigger

Connecting the lattice columns to the core to make an integrated mega-frame cylinder.

Band Truss

Reinforcing the towers around the outermost perimeter and acting as a base for the gravity columns.

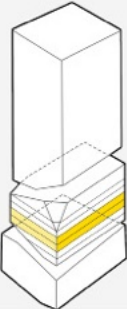
Hanging Pillar

Suspended from ring belt trusses and reach trusses.

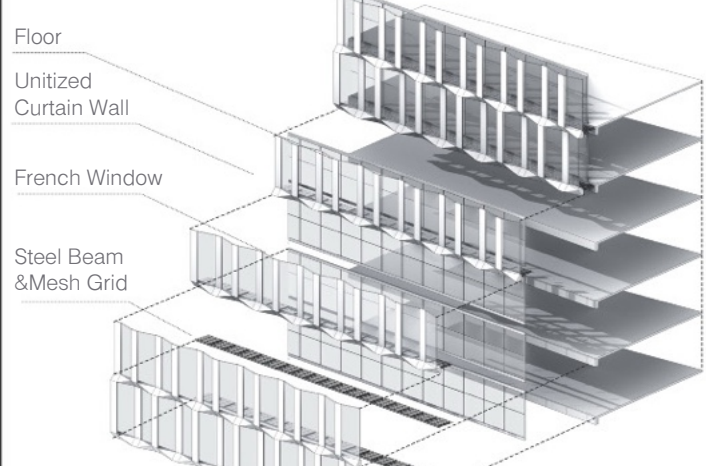
4M X 5M Square Lattice Column

A giant lattice column formed by four small 1m X 1m columns tied to each other.

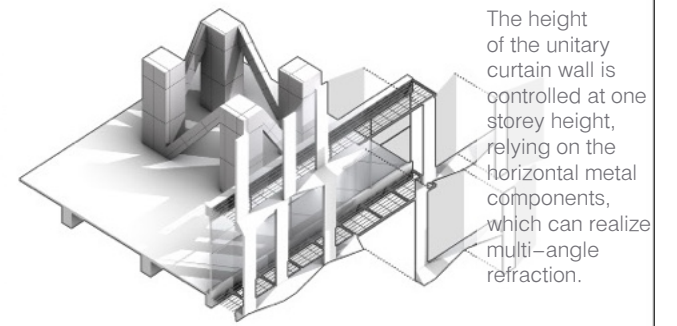
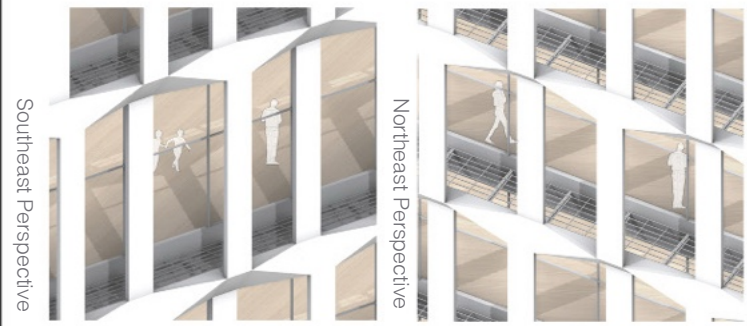
Schematic Location of Structural Details



CURTAIN WALL EXPLODED VIEW



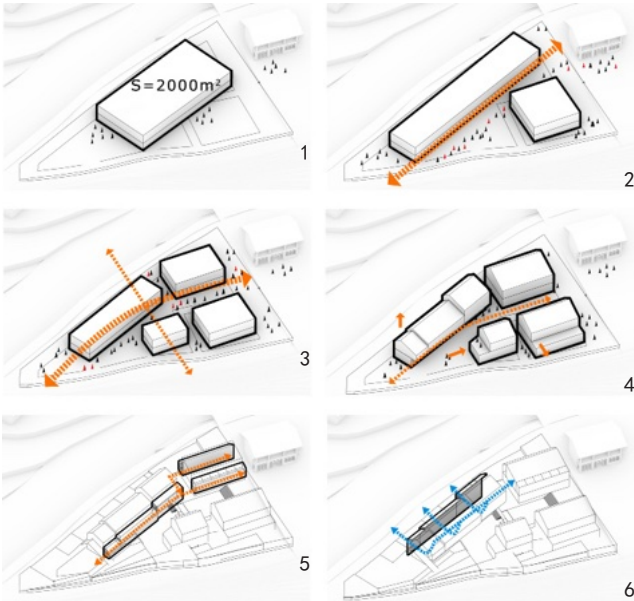
The building uses a unitized curtain wall that is disconnected from the main body of the building: floor-to-ceiling glass on the inside, and curtain wall at different angles on the outside, connected to the floor slabs by steel beams and mesh grids. The curtain wall units are made of highly reflective glass, arrayed to give a different sense of order at different angles.



The height of the unitary curtain wall is controlled at one storey height, relying on the horizontal metal components, which can realize multi-angle refraction.

FALLING INTO THE VALLEY

VOLUME GENERATION



DESIGN DESCRIPTION

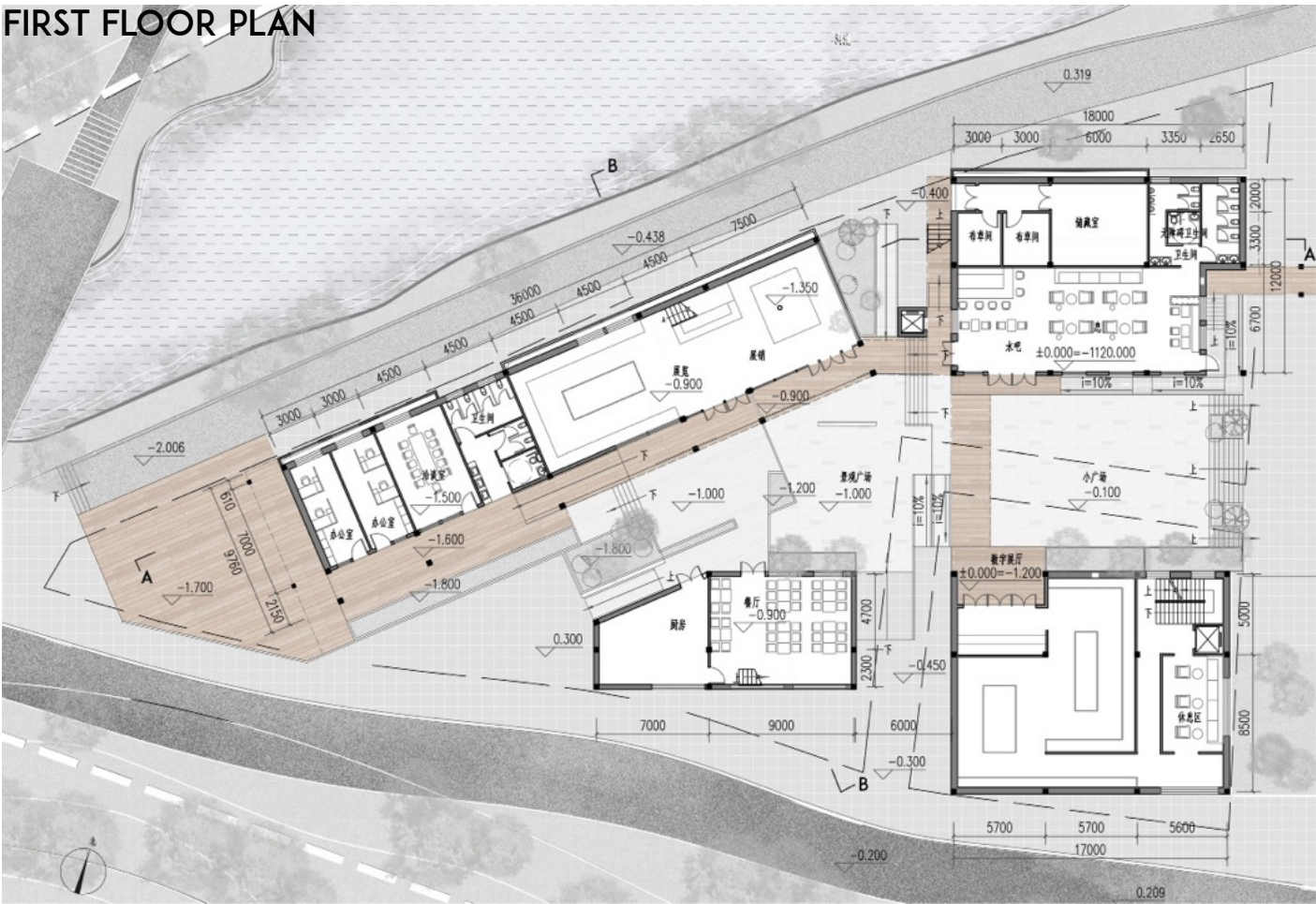
Pingwu County in Sichuan Province is the settlement of the Baima Tibetan people. The design is inspired by the plumes on the hats of the Baima Tibetans, which are translated into the architectural framework and layout. Taking into account the local climate, which is cold in winter and cool in summer, the building combines a sun room, solar glass panels, ecological water circulation system, double cavity walls, etc. with the roof and facade to improve the utilization of solar energy as much as possible. Low-carbon materials and local materials are used to reduce carbon emissions and improve the eco-efficiency of the building.

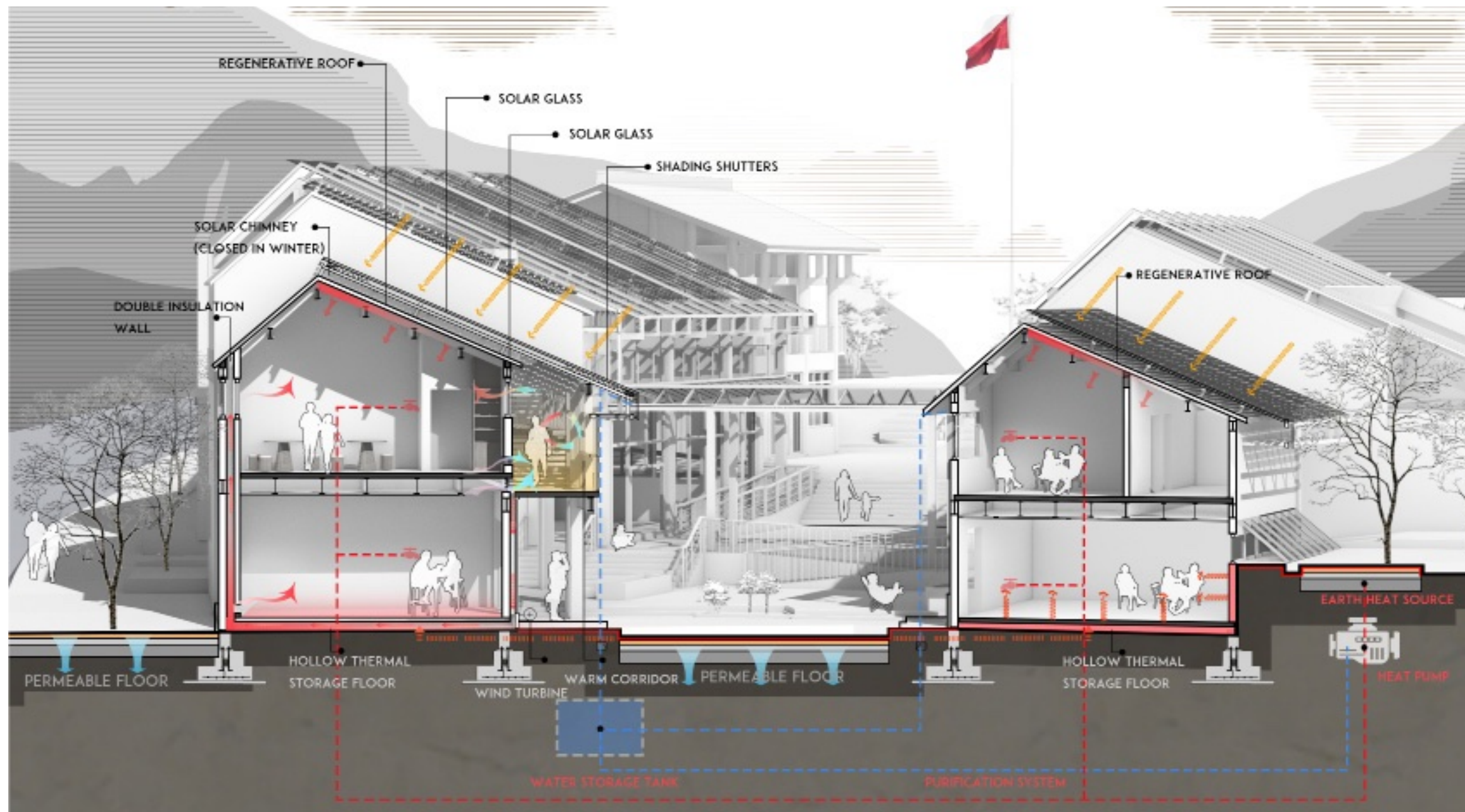


MASTER FLOOR PLAN

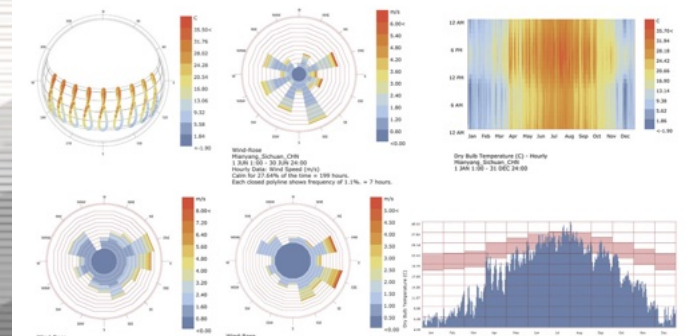


FIRST FLOOR PLAN



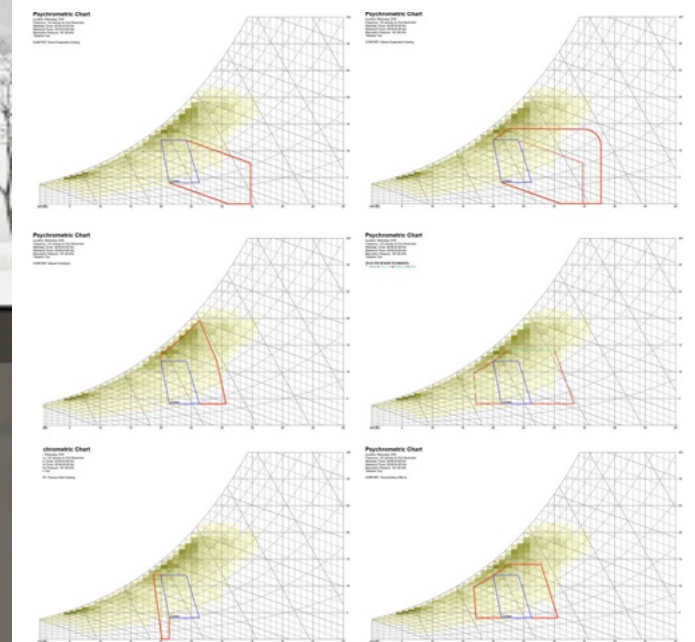


CLIMATIC SIMULATION



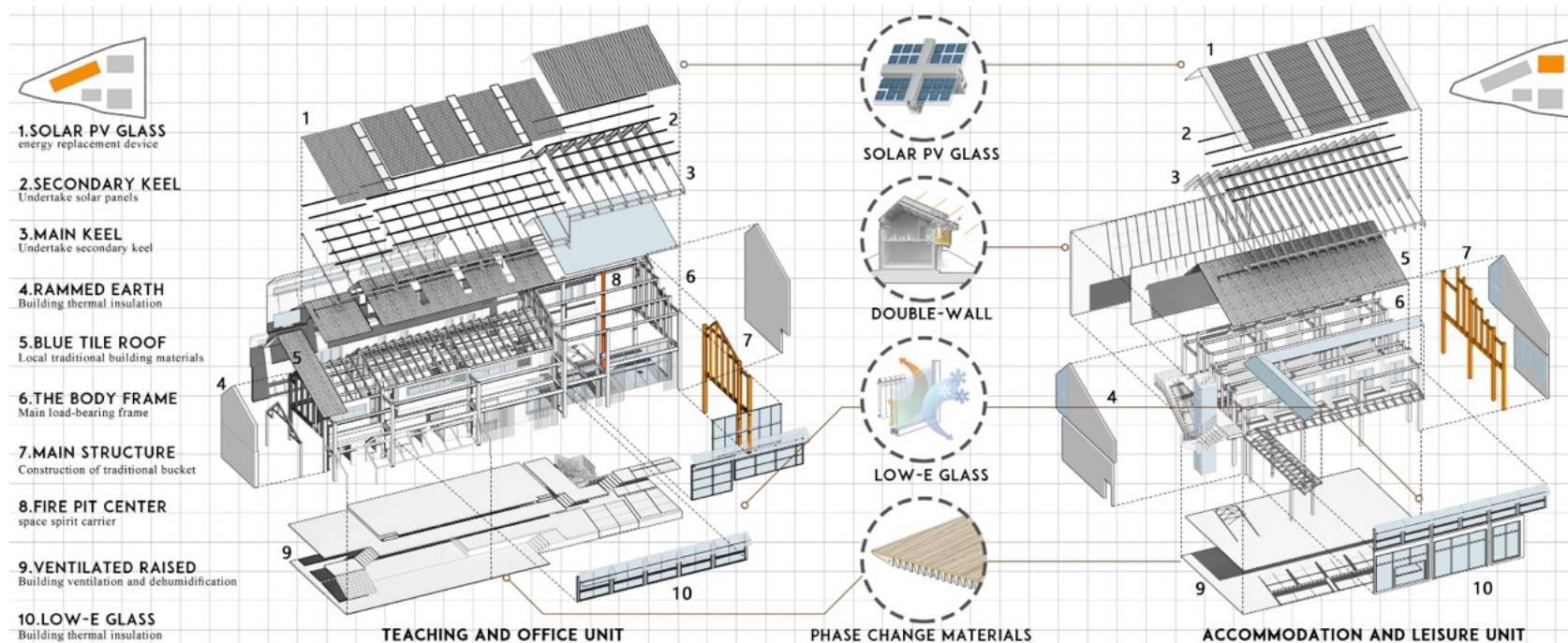
Cool and rainy summers, cold and snowy winters, average annual rainfall 800mm.

ENTHALPY AND HUMIDITY DIAGRAM

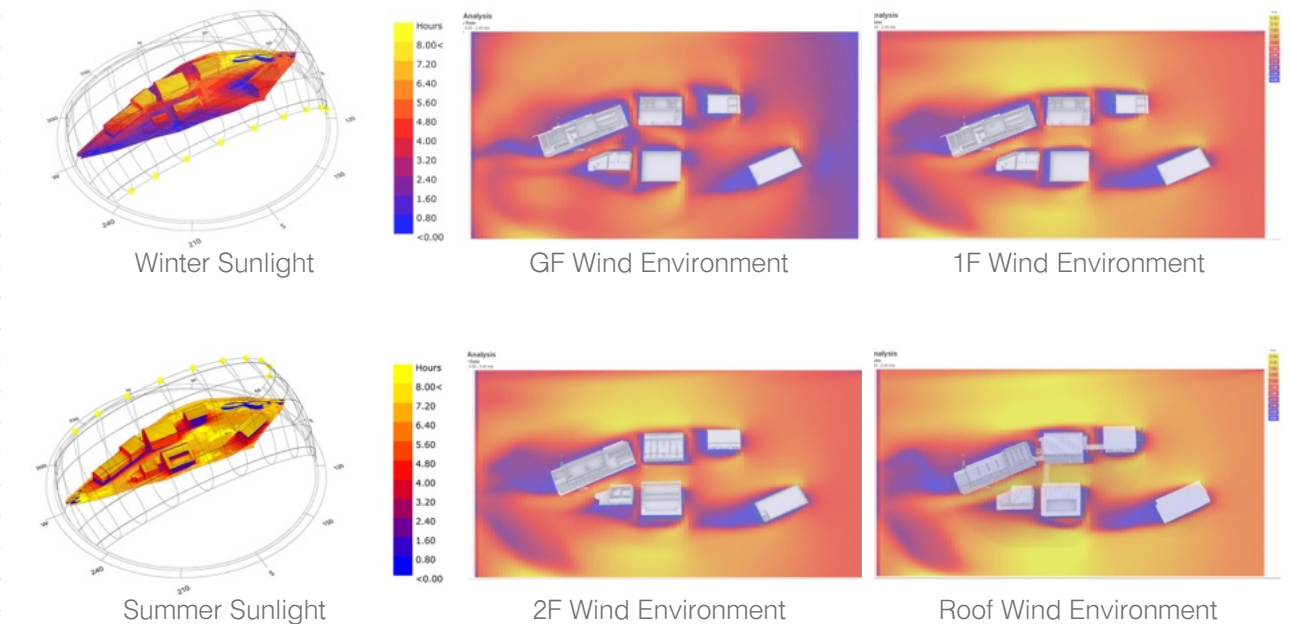


The design is well ventilated and lighted, with appropriate passive and active technologies.

EXPLODED VIEW



WIND ENVIRONMENT SIMULATION

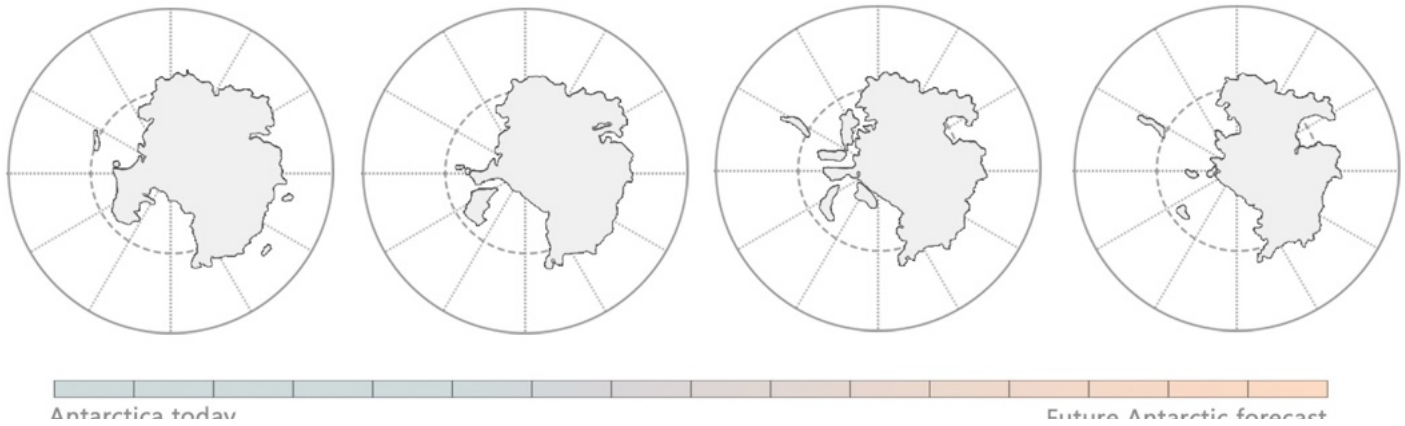
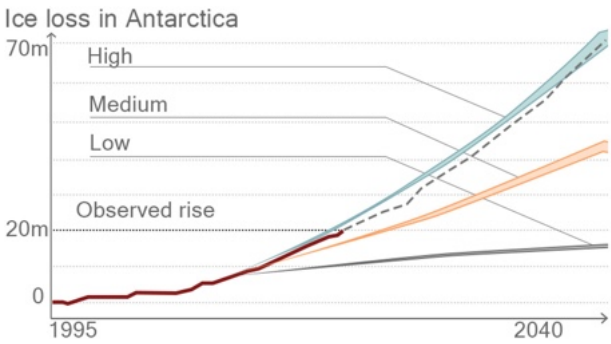


The impact of morning and evening valley winds on the ventilation and energy efficiency of the building was taken into account in the layout of the building blocks, resulting in good ventilation conditions in the internal corridors of the building.

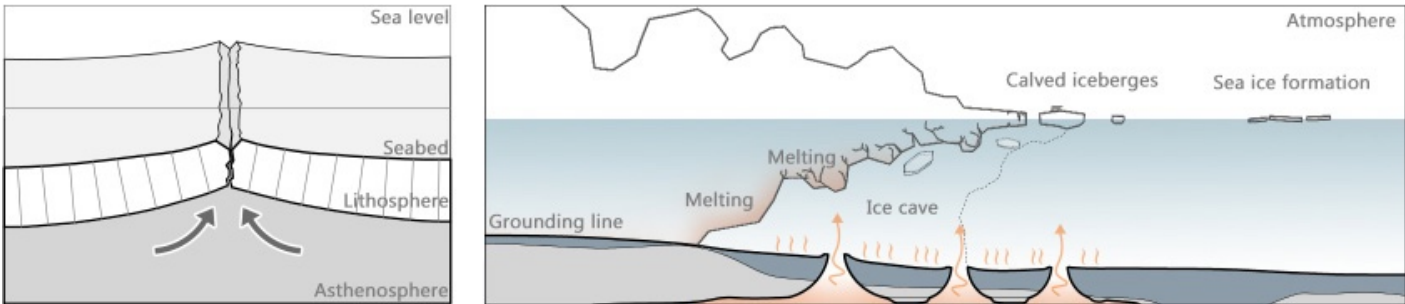
ANTARCTIC RESCUE LINE

MELTING OF ANTARCTIC GLACIERS

According to the US National Snow and Ice Data Center(NSIDC), Antarctic sea ice shrank to 1.91 million square kilometers in 2023, the lowest level since records began in 1979. The continent’s glaciers are also melting faster. After the sea ice completely melts, the continent’s glaciers are also in danger. In our study of the melting of the Antarctic ice, we found that the West Antarctic ice is melting at a much faster rate than the east.



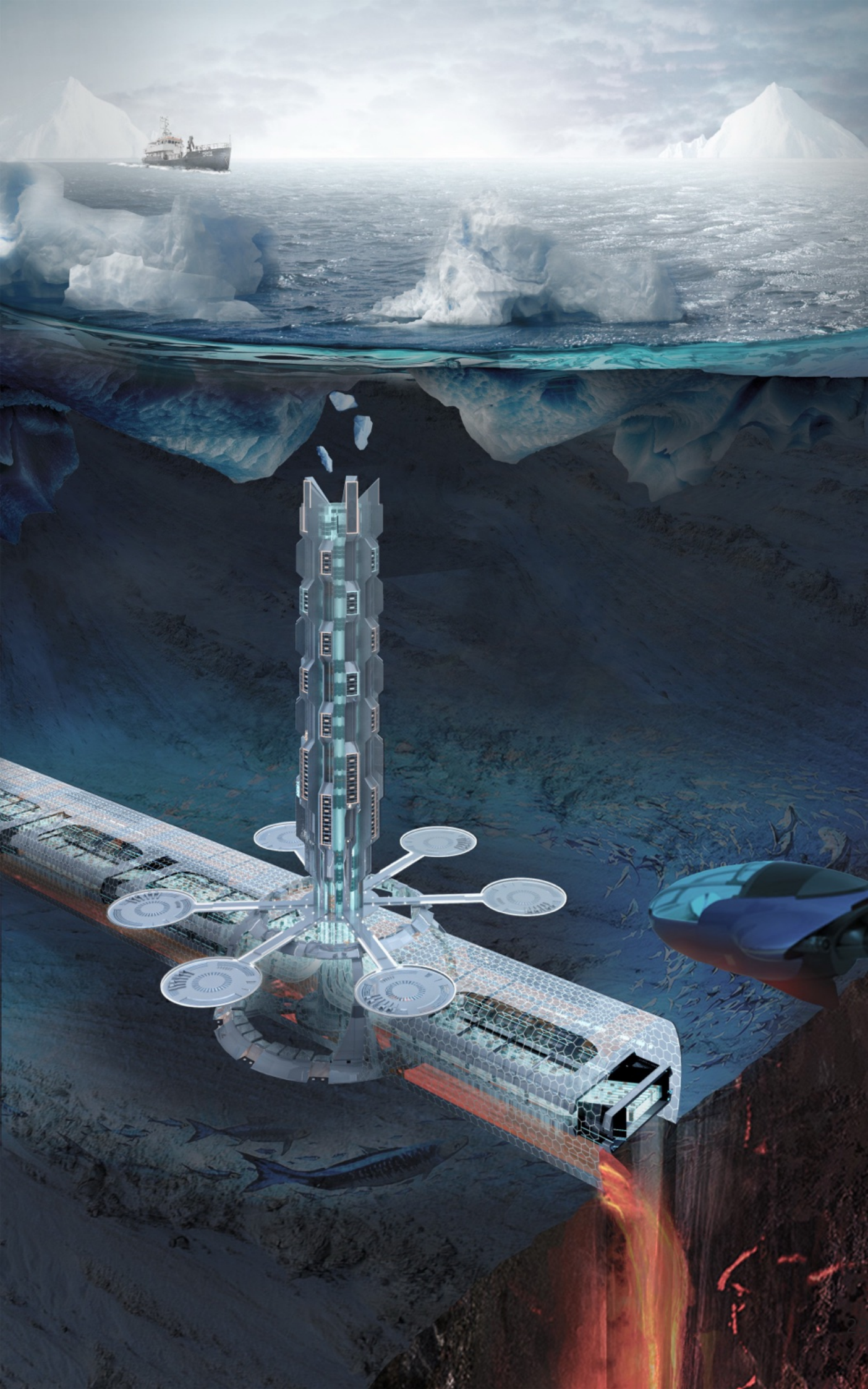
CONCEPT DEVELOPMENT

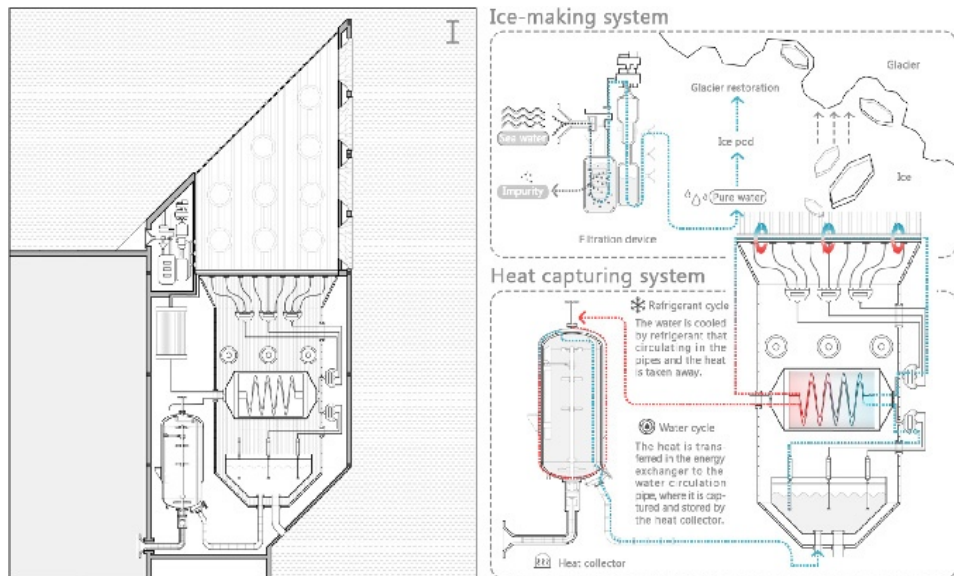


To minimize the impact of geothermal radiation on glaciers, we plan to block the heat from its source: cracks in the earth’s crust. Because the cracks in the Earth’s crust are so long, we took inspiration for our skyscraper from THE LINE, which is going to be built in Saudi Arabia. Our skyscraper can extend indefinitely depending on the length of the crack, and has the function of converting heat into electricity and making ice.

Our skyscrapers would absorb the heat from the cracks and convert it into electricity, which would then be charged in battery cells. Battery-carrying submarines from other areas will transport the empty batteries here and the full batteries to other areas for use.

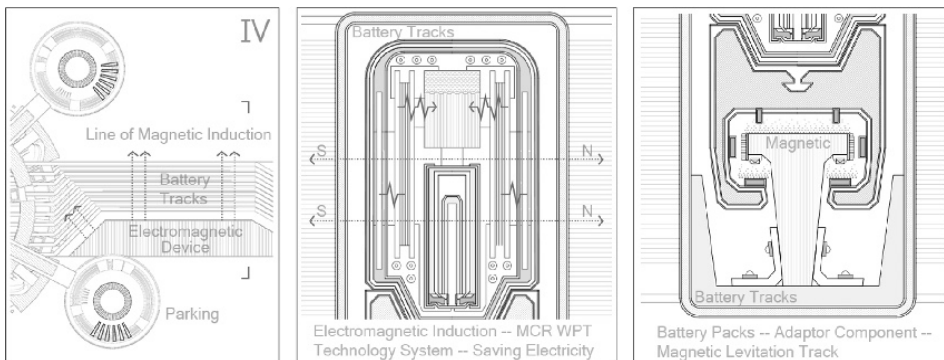
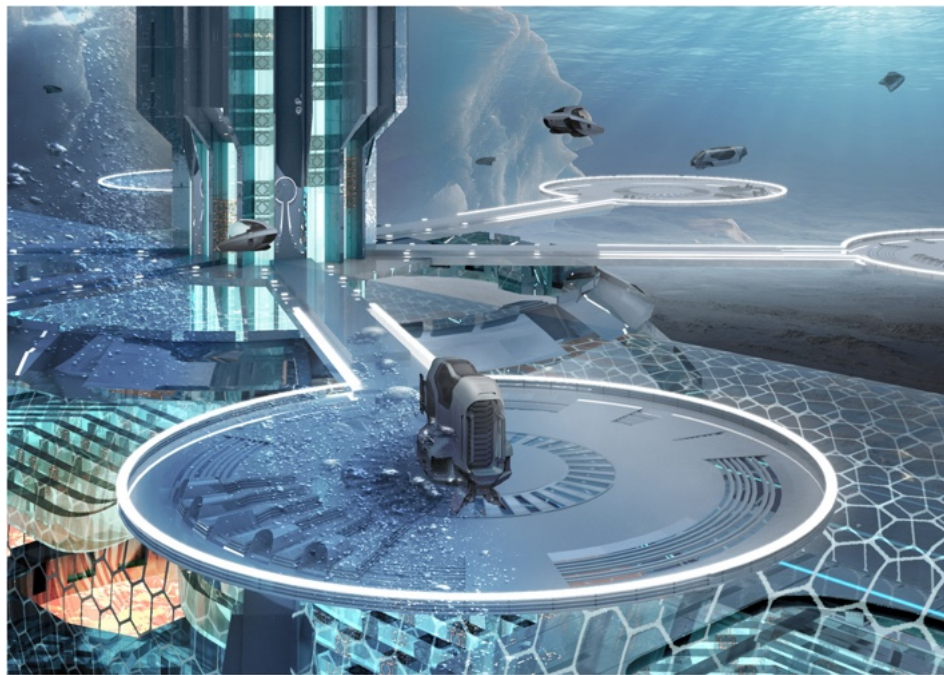
In addition, the crown part of our building tower has the function of making ice. The ice created by the building rises upward due to buoyancy. The ice created by skyscrapers in the ice cave will gradually accumulate on the top walls of the cave, strengthening and restoring the glacier. The ice created by skyscrapers outside the caves floats to the surface, increasing the extent of sea ice.





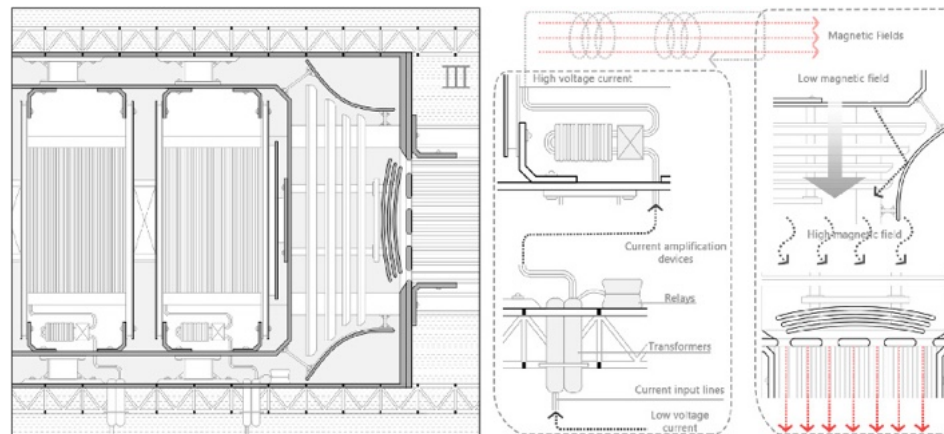
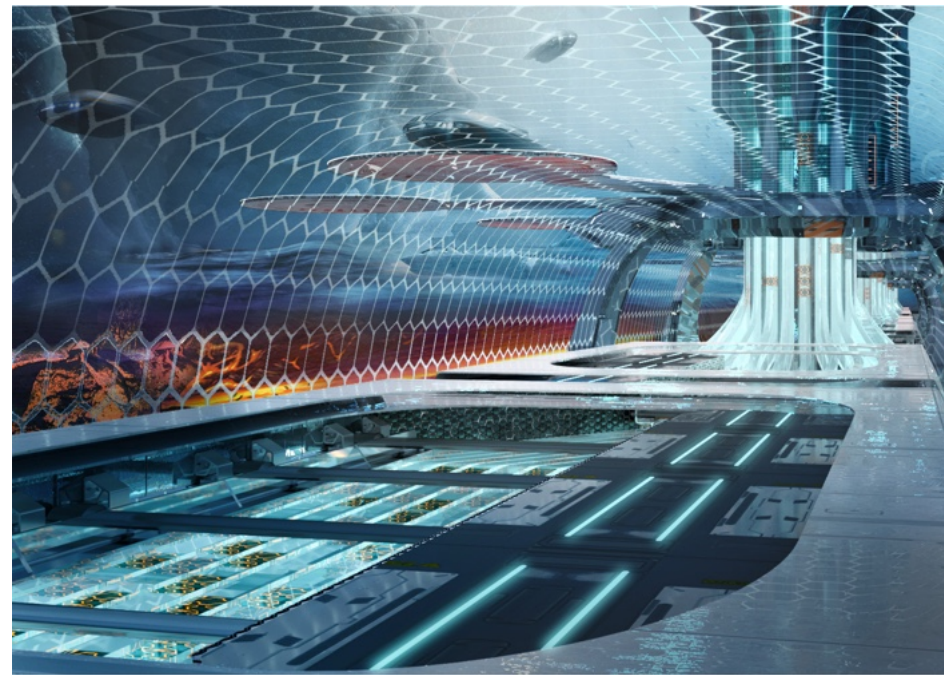
There are several battery tunnels between the two adjacent skyscrapers. When the submarine is docked, small battery-carrying robots take the empty batteries out of it and send them through a battery portal in the skyscraper into a battery tunnel inside.

The empty battery continues to travel along the track to the top of the skyscraper, where it switches tracks on the top loop, and then moves down from another track adjacent to the original track to the bottom horizontal position for charging. As the battery cell moves along the bottom horizontal area, it gradually becomes charged due to electromagnetic induction. The full cell then travels to an adjacent building where it is stored, waiting for a battery-carrying robot to transport it to the submarine.

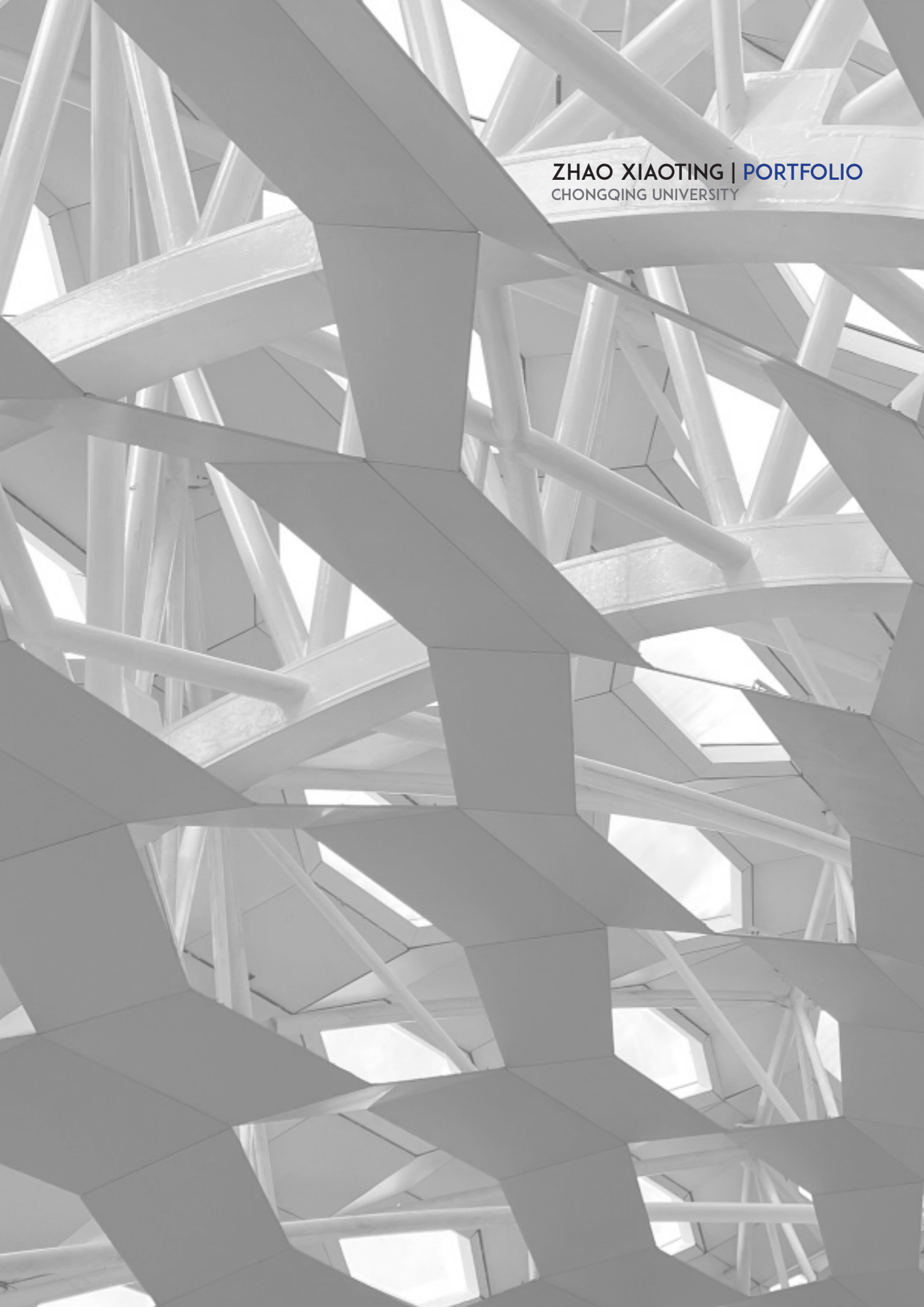


The battery portal on the skyscrapers has an indicator light that switches between orange and blue colors. When the indicator is orange, the battery warehouse inside the portal is full of fully charged battery cells. If the indicator is blue, the battery warehouse is not fully stocked. The battery-carrying robot automatically recognize the color of the indicator light to determine which portal to take the battery from.

Each battery cell does not emit light in the absence of electricity. As the battery moves along the battery track to charge, the cell will gradually change from a non-glowing state to a glowing state. The brightness of a cell's light indicates how much energy is stored inside.



Heat from cracks in the earth's crust is absorbed through thermoelectric conversion system at the base of the building. The rest of the unabsorbed heat will be trapped by a thermal shield on the outside of the building. The heat generated by the ice-making system is released into the space between the core and the outer space of the building, back to the bottom.



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